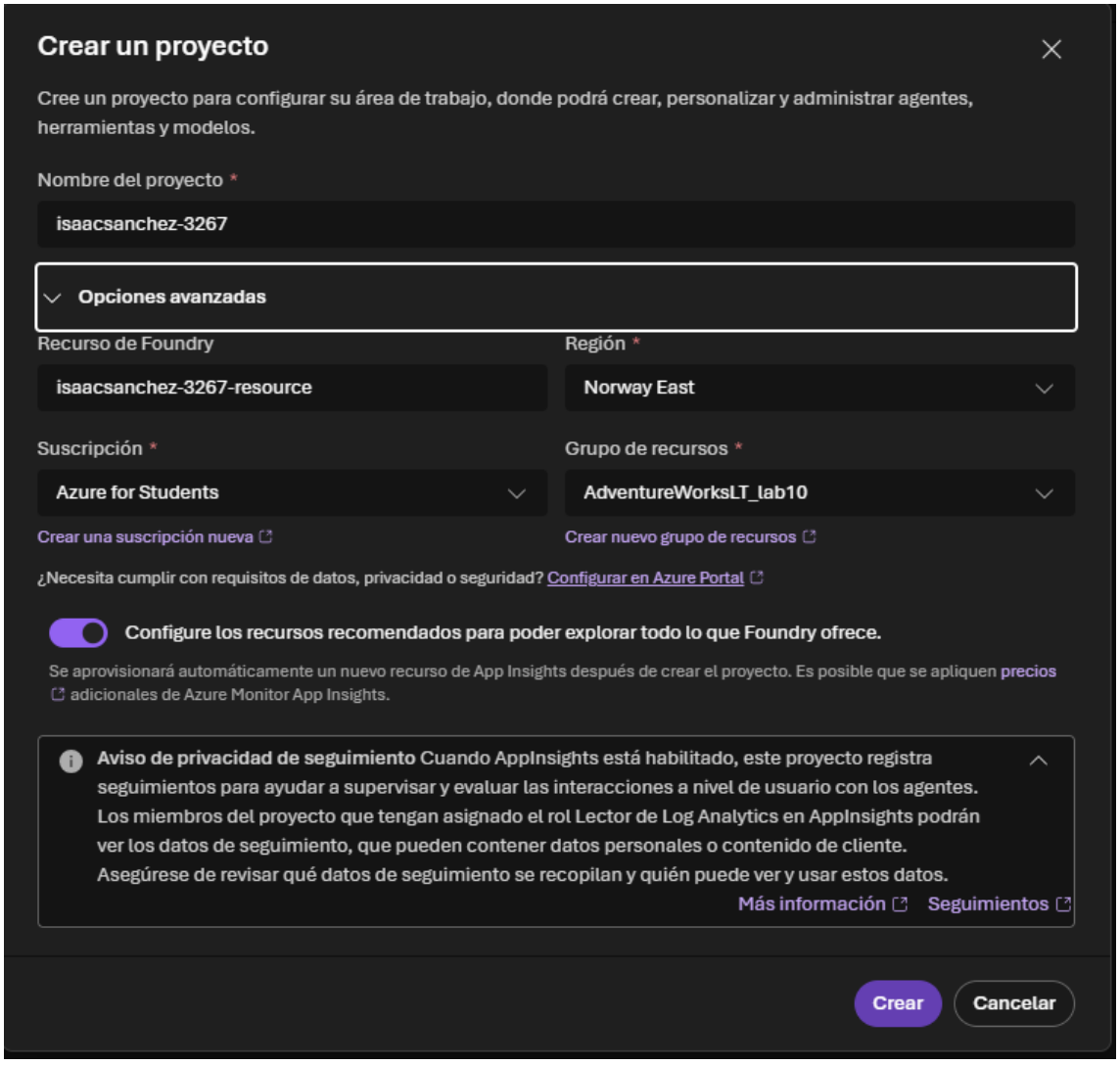
Contenido

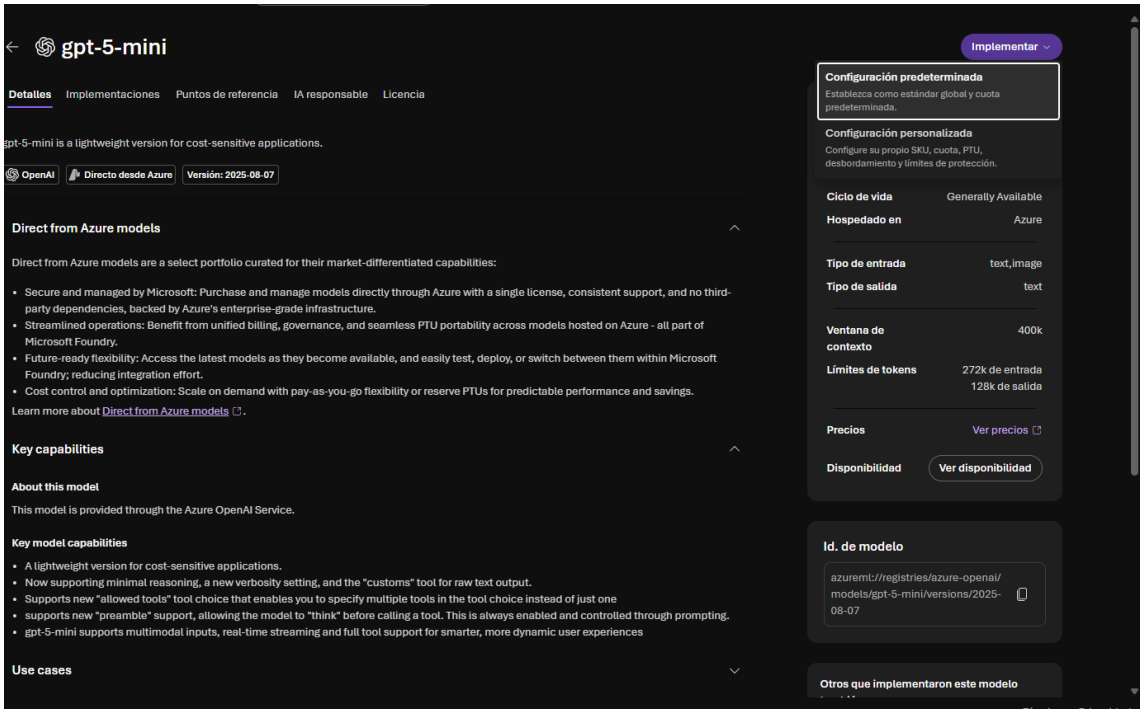
Create a Foundry project and deploy Azure OpenAI models 2
Create a Foundry project 2
Deploy Azure OpenAI models 3
Retrieve the Azure OpenAI endpoint 4
Configure managed identity Access create the ProductReview table 4
Create a database scoped credential and an external model for embeddings 5
Add a vector column and generate embeddings 7
Create a full-text index 9
Search with full-text predicates 10
Search with vector similarity 12
Combine full-text and vector search with hybrid search 14
Compare the three search approaches 14

Create a Foundry project and deploy Azure OpenAI models

Create a Foundry project

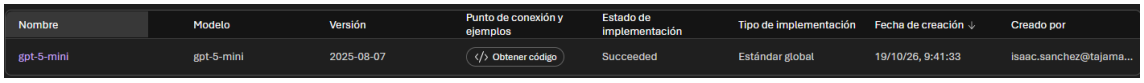


Deploy Azure OpenAI models

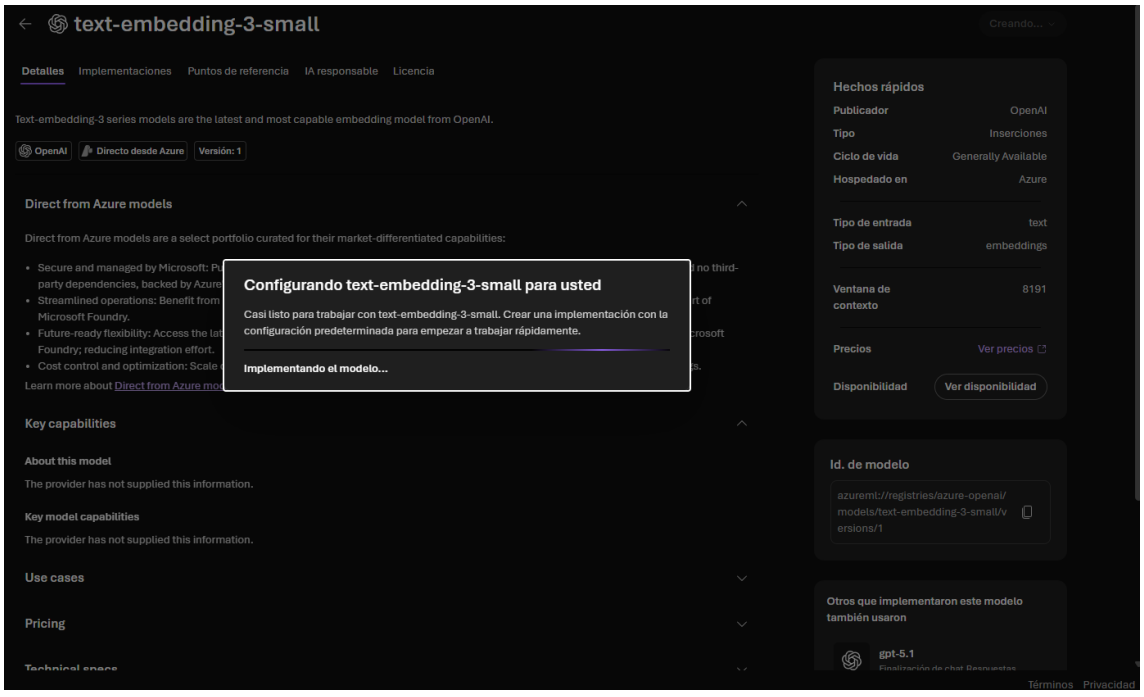


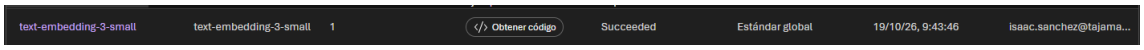
Utilizamos gpt5mini

Y creamos el modelo

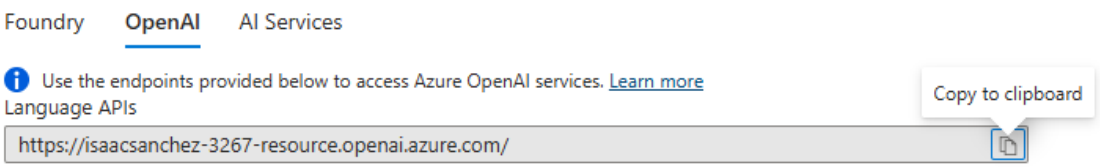


Repetimos con el text-embedding





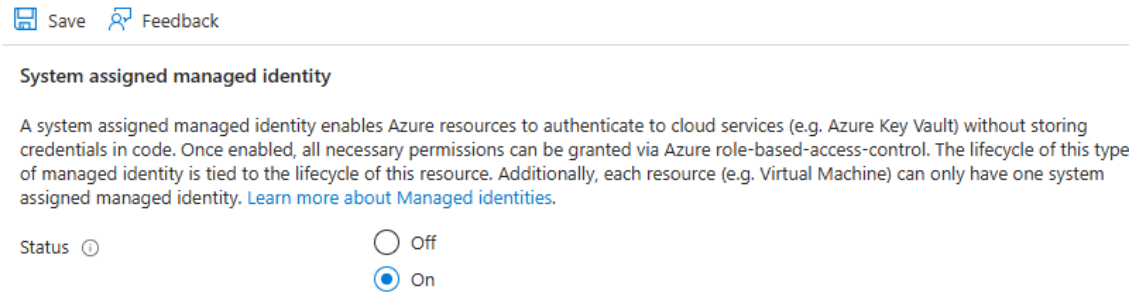
Retrieve the Azure OpenAI endpoint



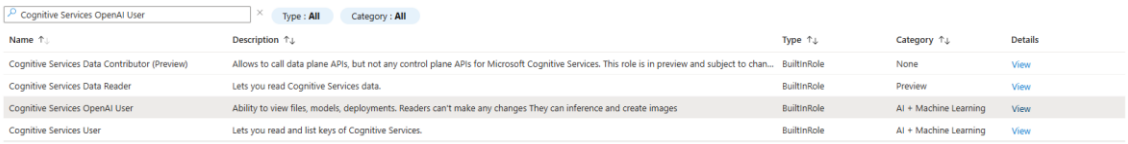
Configure managed identity Access create the ProductReview table

Vamos al servidor de la bbdd

En la prte de seguridad activamos managed identityen



En el menú de acceso token agregamos un rol que sea Cognitive Services OpenAI User

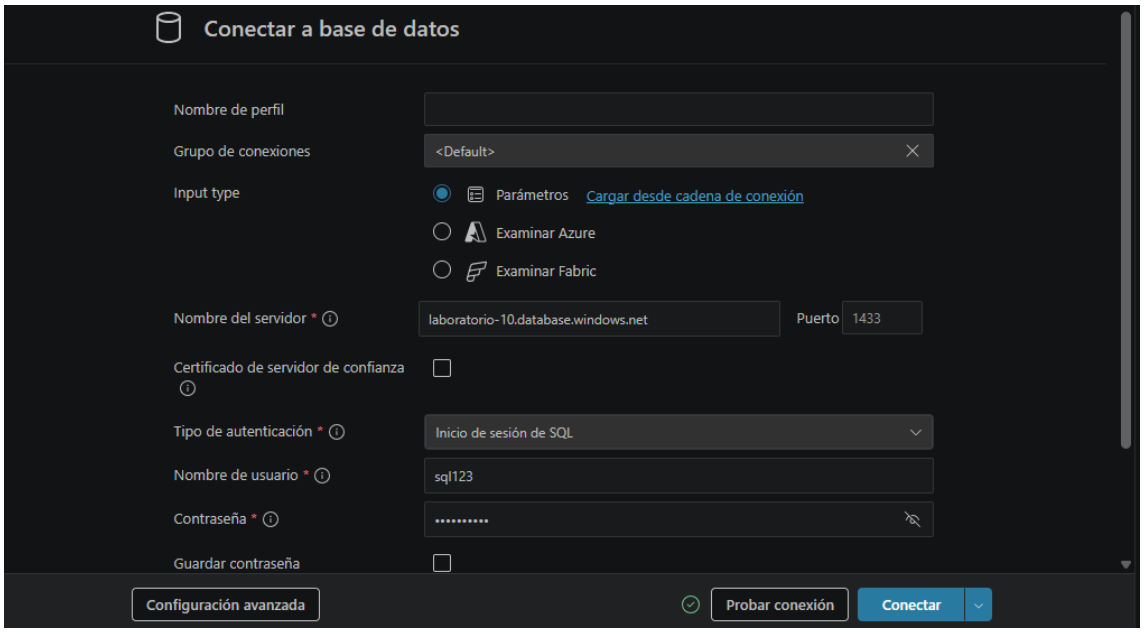


Asignamos el sql server como miembro

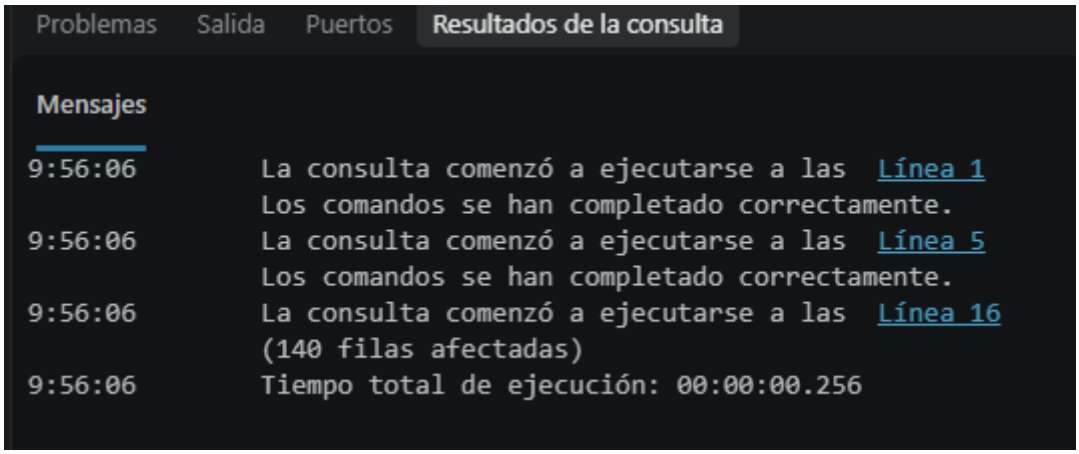


Create a database scoped credential and an external model for embeddings

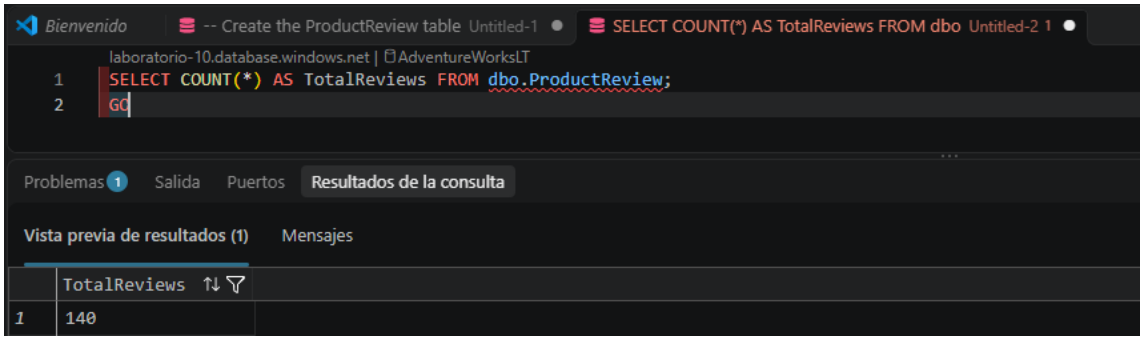
Mediante visual estudio nos conctamos a la bbdd que hemos creado



Una vez conectados ejecutamos el archivo para la creación de la bbdd



Comprobación de que la creación a sido exitosa



Ejecutamos este código

```
-- Create a database master key if one doesn't exist
 IF NOT EXISTS (SELECT * FROM sys.symmetric_keys WHERE name =
'##MS_DatabaseMasterKey##')
     CREATE MASTER KEY ENCRYPTION BY PASSWORD = '<your strong password
here>';
 GO

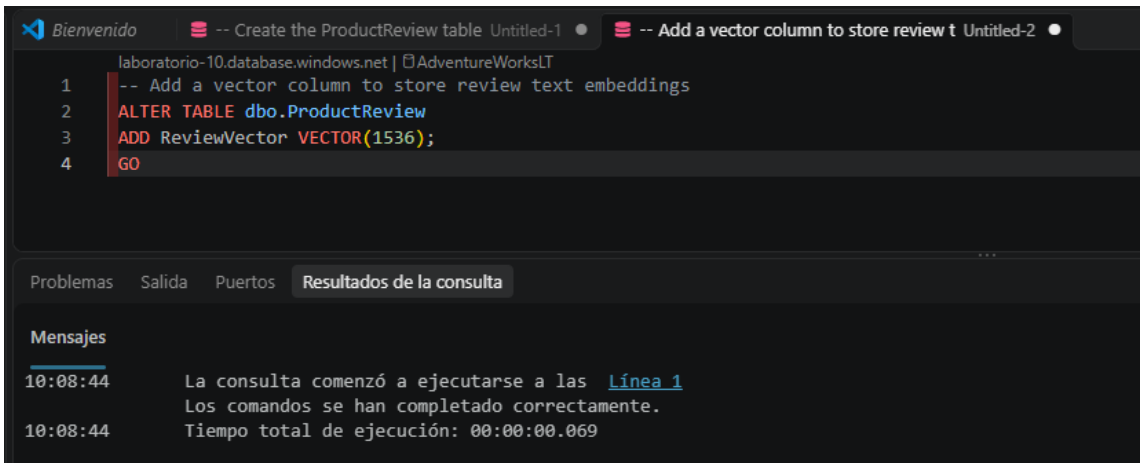
 -- Create a credential using Managed Identity
 CREATE DATABASE SCOPED CREDENTIAL [https://<your-openai-
endpoint>.openai.azure.com]
 WITH IDENTITY = 'Managed Identity', SECRET =
'{"resourceid":"https://cognitiveservices.azure.com"}';
 GO
```

Creamos el modelo externo

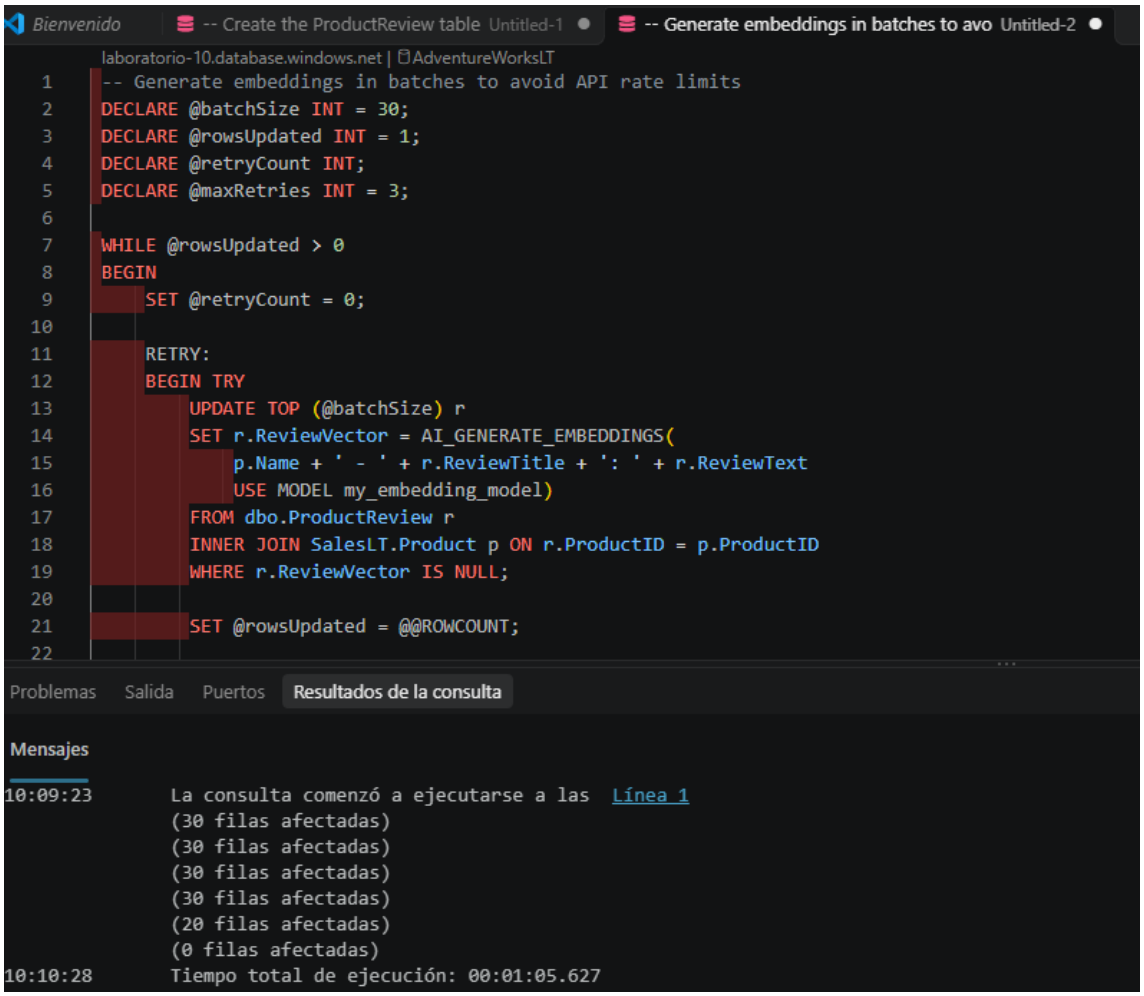


Add a vector column and generate embeddings

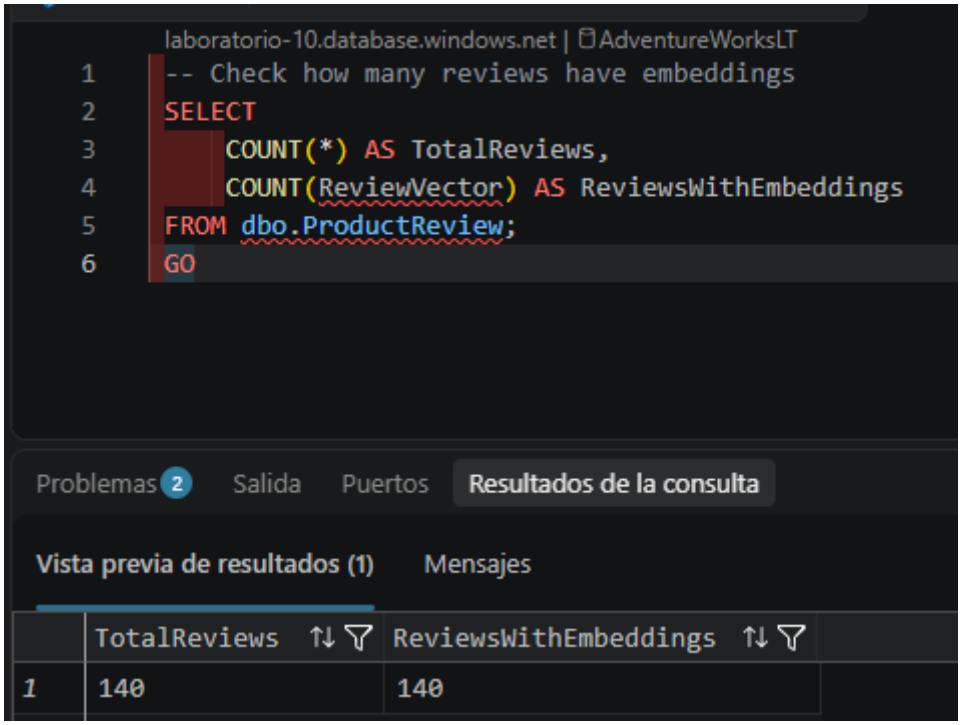
Añadimos la columna de embeddins



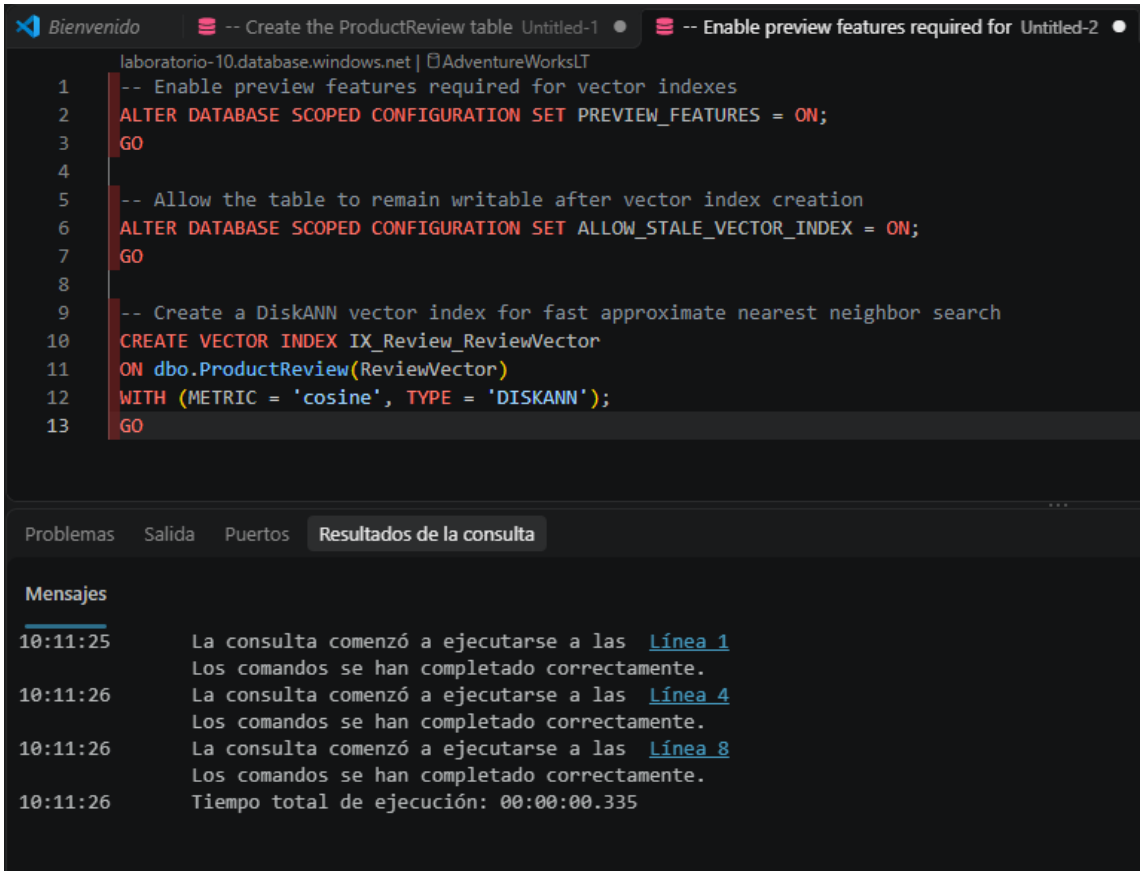
Generamos los embeddings



Comprobamos los embeddings creados

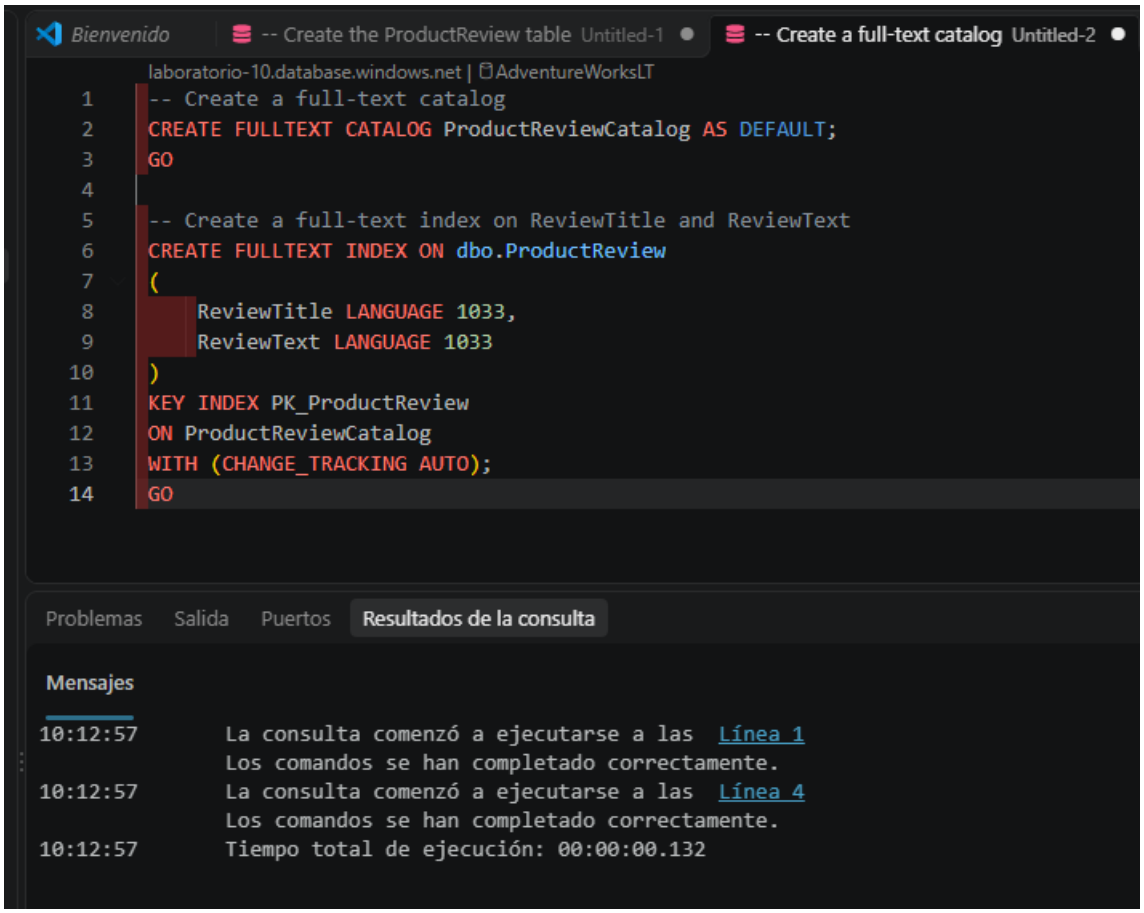


Creamos el índice de los vectores



Create a full-text index

Creamos el catálogo completo

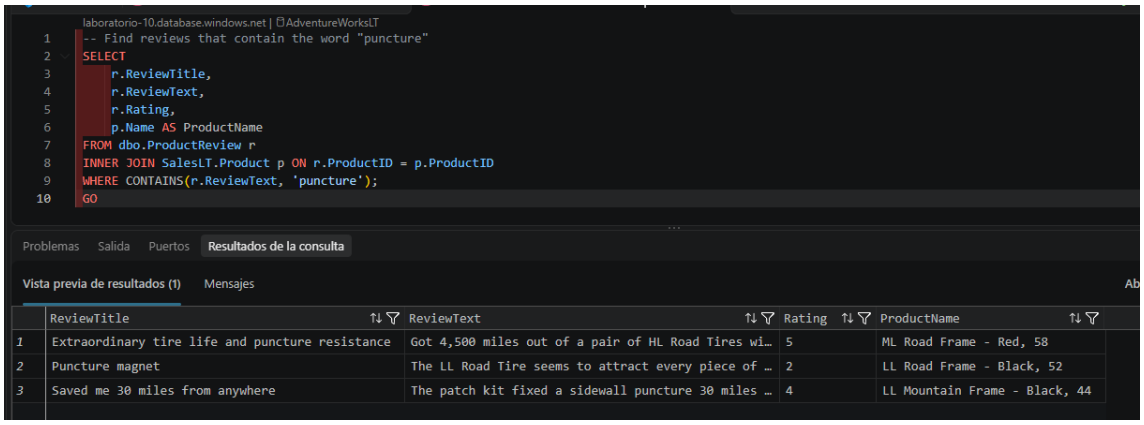


Verificamos la creación de los índices



Search with full-text predicates

Buscamos por una palabra concreta



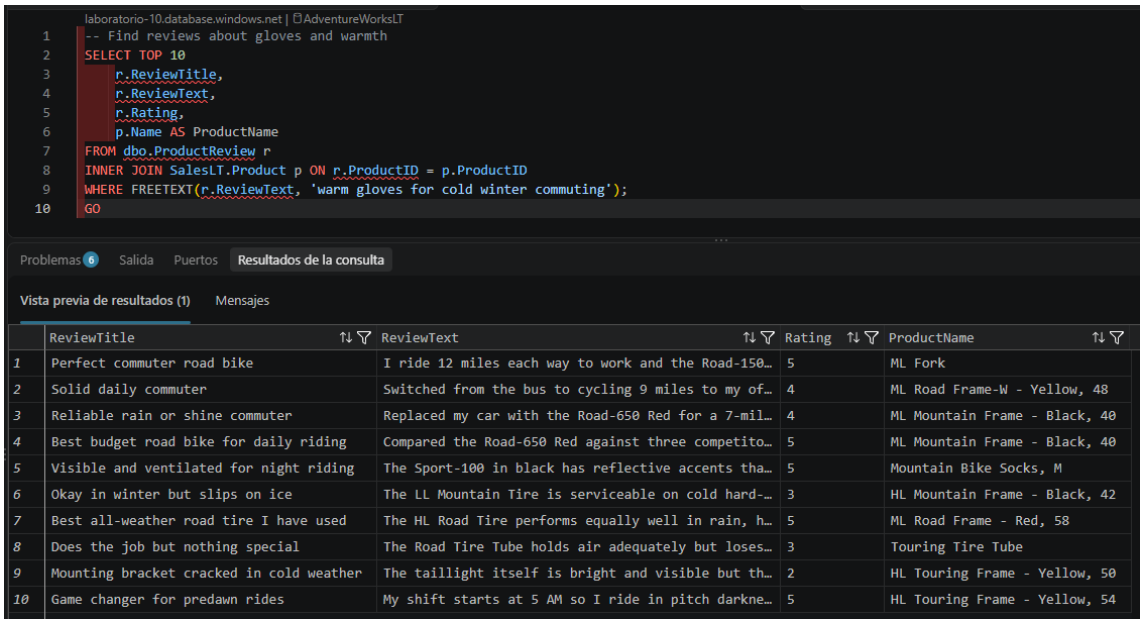
```
laboratorio-10.database.windows.net | AdventureWorksLT
-- Find reviews that contain the word "puncture"
SELECT
    r.ReviewTitle,
    r.ReviewText,
    r.Rating,
    p.Name AS ProductName
FROM dbo.ProductReview r
INNER JOIN SalesLT.Product p ON r.ProductID = p.ProductID
WHERE CONTAINS(r.ReviewText, 'puncture');
GO
```

Problemas Salida Puertos Resultados de la consulta

Vista previa de resultados (1) Mensajes

	ReviewTitle	ReviewText	Rating	ProductName
1	Extraordinary tire life and puncture resistance	Got 4,500 miles out of a pair of HL Road Tires wi…	5	ML Road Frame - Red, 58
2	Puncture magnet	The LL Road Tire seems to attract every piece of …	2	LL Road Frame - Black, 52
3	Saved me 30 miles from anywhere	The patch kit fixed a sidewall puncture 30 miles …	4	LL Mountain Frame - Black, 44

Buscamos por una frase



```
laboratorio-10.database.windows.net | AdventureWorksLT
-- Find reviews about gloves and warmth
SELECT TOP 10
    r.ReviewTitle,
    r.ReviewText,
    r.Rating,
    p.Name AS ProductName
FROM dbo.ProductReview r
INNER JOIN SalesLT.Product p ON r.ProductID = p.ProductID
WHERE FREETEXT(r.ReviewText, 'warm gloves for cold winter commuting');
GO
```

Problemas 6 Salida Puertos Resultados de la consulta

Vista previa de resultados (1) Mensajes

	ReviewTitle	ReviewText	Rating	ProductName
1	Perfect commuter road bike	I ride 12 miles each way to work and the Road-150…	5	ML Fork
2	Solid daily commuter	Switched from the bus to cycling 9 miles to my of…	4	ML Road Frame-W - Yellow, 48
3	Reliable rain or shine commuter	Replaced my car with the Road-650 Red for a 7-mil…	4	ML Mountain Frame - Black, 40
4	Best budget road bike for daily riding	Compared the Road-650 Red against three competito…	5	ML Mountain Frame - Black, 40
5	Visible and ventilated for night riding	The Sport-100 in black has reflective accents tha…	5	Mountain Bike Socks, M
6	Okay in winter but slips on ice	The LL Mountain Tire is serviceable on cold hard-…	3	HL Mountain Frame - Black, 42
7	Best all-weather road tire I have used	The HL Road Tire performs equally well in rain, h…	5	ML Road Frame - Red, 58
8	Does the job but nothing special	The Road Tire Tube holds air adequately but loses…	3	Touring Tire Tube
9	Mounting bracket cracked in cold weather	The taillight itself is bright and visible but th…	2	HL Touring Frame - Yellow, 50
10	Game changer for predawn rides	My shift starts at 5 AM so I ride in pitch darkne…	5	HL Touring Frame - Yellow, 54

Ahora utilizamos constaintable para obtener las puntuaciones de similitud

```
laboratorio-10.database.windows.net | AdventureWorksLT
-- Search with ranking scores
SELECT TOP 10
    r.ReviewTitle,
    r.ReviewText,
    r.Rating,
    p.Name AS ProductName,
    ft.[RANK] AS FullTextRank
FROM dbo.ProductReview r
INNER JOIN SalesLT.Product p ON r.ProductID = p.ProductID
INNER JOIN CONTAINSTABLE(dbo.ProductReview, (ReviewTitle, ReviewText),
    'FORMSOF(INFLECTIONAL, mountain) AND FORMSOF(INFLECTIONAL, trail)') AS ft
    ON r.ReviewID = ft.[KEY]
ORDER BY ft.[RANK] DESC;
```

Problemas 2 | Salida | Puertos | Resultados de la consulta

Vista previa de resultados (1) | Mensajes | Abrir en nueva pestaña

	ReviewTitle	ReviewText	Rating	ProductName	FullTextRank
1	Perfect weekend trail companion	We take the Mountain-500 Silver out every Saturda…	5	Road-250 Black, 52	21
2	Decent starter but has limits	The Mountain-500 Black is fine for fire roads and…	3	Road-550-W Yellow, 42	21
3	Grip for days on loose terrain	Switched to the HL Mountain Tire and the differen…	5	HL Mountain Frame - Silver, 42	10
4	Amazing off-road but noisy on pavement	The HL Mountain Tire has incredible trail grip bu…	4	HL Mountain Frame - Silver, 42	10
5	Okay in winter but slips on ice	The LL Mountain Tire is serviceable on cold hard-…	3	HL Mountain Frame - Black, 42	10
6	Platform pedals done right	The HL Mountain Pedals have the grip and durabili…	4	ML Mountain Pedal	10
7	Rugged mountain bottle cage	The Mountain Bottle Cage holds my oversized insul…	4	Road Bottle Cage	10
8	Virtually indestructible	The Mountain Bottle Cage has been on my bike thro…	5	Road Bottle Cage	10
9	Compact and surprisingly effective	The Minipump is small enough to carry in a jersey…	4	HL Touring Frame - Yellow, 46	10
10	Durability after years of hard riding	Three years and 8,000 miles of aggressive trail r…	5	Mountain-100 Black, 38	10

Ahora buscamos por el inicio de una palabra

```
laboratorio-10.database.windows.net | AdventureWorksLT
-- Find reviews with words starting with "comfort"
SELECT
    r.ReviewTitle,
    r.ReviewText,
    r.Rating,
    p.Name AS ProductName
FROM dbo.ProductReview r
INNER JOIN SalesLT.Product p ON r.ProductID = p.ProductID
WHERE CONTAINS(r.ReviewText, '"comfort*"');
GO
```

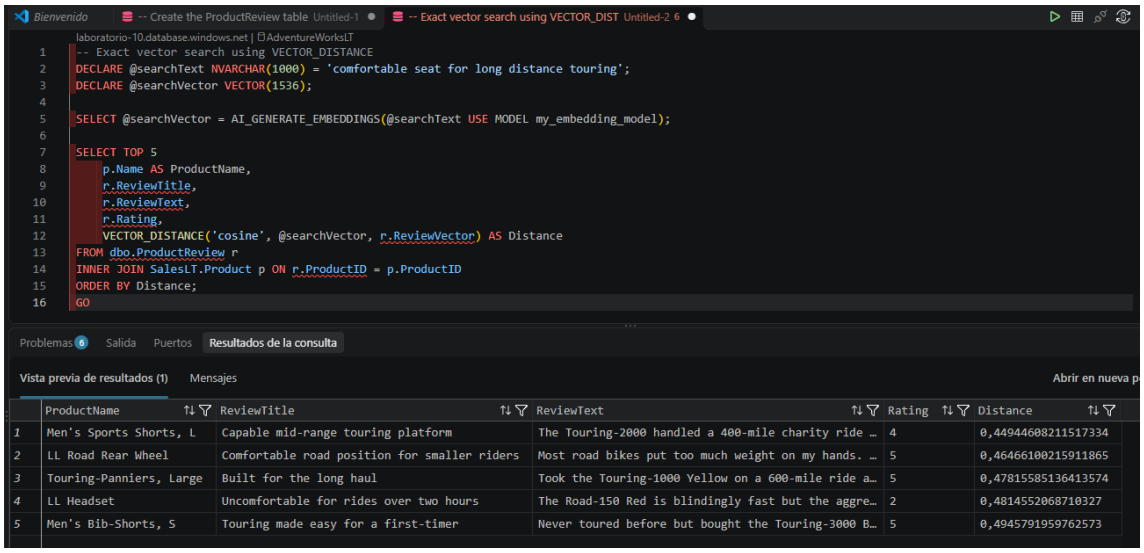
Problemas 6 | Salida | Puertos | Resultados de la consulta

Vista previa de resultados (1) | Mensajes

	ReviewTitle	ReviewText	Rating	ProductName
1	Sunday ride perfection	The Road-550-W is my go-to for Sunday morning rid…	5	HL Mountain Rear Wheel
2	Handlebars flex during hard efforts	Comfortable for cruising, but when I stand and sp…	3	LL Road Rear Wheel
3	Great first road bike	Transitioned from a hybrid to the Road-650 Black …	4	ML Road Frame-W - Yellow, 48
4	Proportions that actually make sense	The Road-350-W has the right proportions for smal…	5	ML Mountain Front Wheel
5	Fit that works for smaller riders	The Mountain-400-W solved every fit issue I had w…	4	Road-250 Black, 44
6	Great family weekend bike	Bought the Mountain-500 Black for Saturday rides …	4	Road-550-W Yellow, 44
7	Fine for paved paths but limited on gravel	The Touring-3000 is comfortable on smooth surface…	3	Women's Tights, L
8	Touring made easy for a first-timer	Never toured before but bought the Touring-3000 B…	5	Men's Bib-Shorts, S
9	Budget touring done right	The Touring-3000 Blue delivered a comfortable 500…	5	Men's Bib-Shorts, M
10	Good entry-level touring bike	Bought the Touring-3000 Yellow for a first bike t…	4	Half-Finger Gloves, M
11	Bright and visible for safety	Chose the Sport-100 Red specifically for visibili…	5	Sport-100 Helmet, Red
12	Good winter commuting glove	These Full-Finger Gloves kept me comfortable on r…	4	Full-Finger Gloves, M
13	Workshop essential for home mechanics	This bike stand makes every maintenance task ten …	5	Short-Sleeve Classic Jersey, S

Search with vector similarity

Usamos vector search para realizar la búsqueda



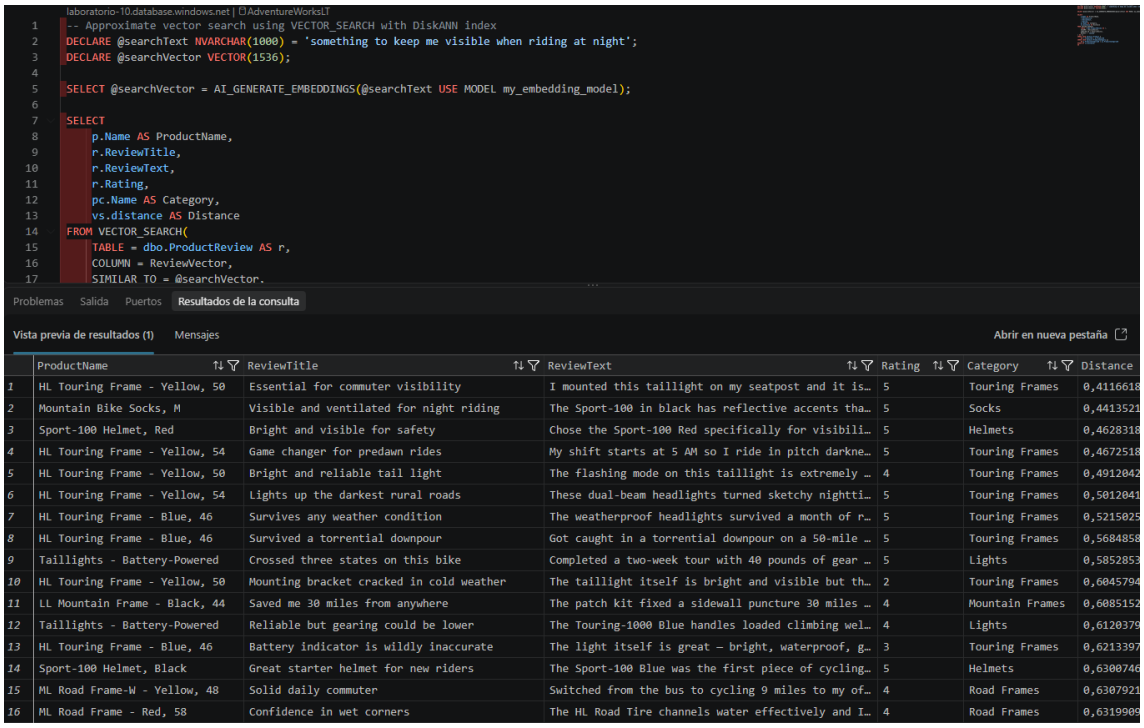
```
-- Exact vector search using VECTOR_DISTANCE
DECLARE @searchText NVARCHAR(1000) = 'comfortable seat for long distance touring';
DECLARE @searchVector VECTOR(1536);

SELECT @searchVector = AI_GENERATE_EMBEDDINGS(@searchText USE MODEL my_embedding_model);

SELECT TOP 5
    p.Name AS ProductName,
    r.ReviewTitle,
    r.ReviewText,
    r.Rating,
    VECTOR_DISTANCE('cosine', @searchVector, r.ReviewVector) AS Distance
FROM dbo.ProductReview r
INNER JOIN SalesLT.Product p ON r.ProductID = p.ProductID
ORDER BY Distance;
GO
```

	ProductName	ReviewTitle	ReviewText	Rating	Distance
1	Men's Sports Shorts, L	Capable mid-range touring platform	The Touring-2000 handled a 400-mile charity ride …	4	0,44944608211517334
2	LL Road Rear Wheel	Comfortable road position for smaller riders	Most road bikes put too much weight on my hands. …	5	0,46466100215911865
3	Touring-Panniers, Large	Built for the long haul	Took the Touring-1000 Yellow on a 600-mile ride a…	5	0,47815585136413574
4	LL Headset	Uncomfortable for rides over two hours	The Road-150 Red is blindingly fast but the aggre…	2	0,4814552068710327
5	Men's Bib-Shorts, S	Touring made easy for a first-timer	Never toured before but bought the Touring-3000 B…	5	0,4945791959762573

Buscamos con el vector y usando el índice para resultados mas rapidos



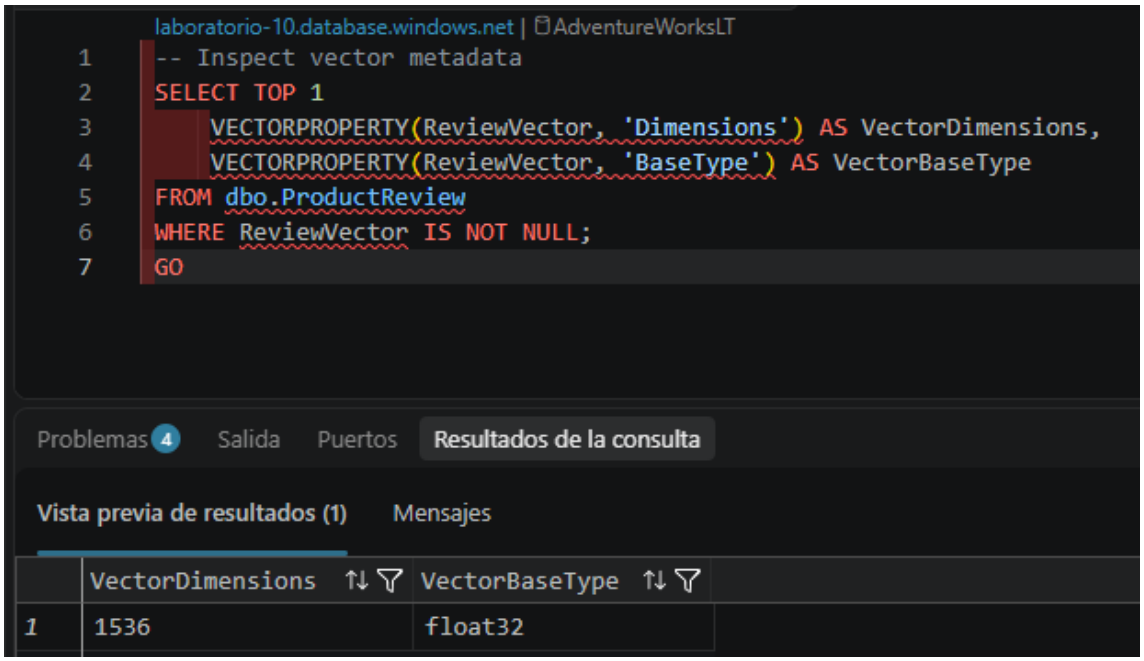
```
-- Approximate vector search using VECTOR_SEARCH with DiskANN index
DECLARE @searchText NVARCHAR(1000) = 'something to keep me visible when riding at night';
DECLARE @searchVector VECTOR(1536);

SELECT @searchVector = AI_GENERATE_EMBEDDINGS(@searchText USE MODEL my_embedding_model);

SELECT
    p.Name AS ProductName,
    r.ReviewTitle,
    r.ReviewText,
    r.Rating,
    pc.Name AS Category,
    vs.distance AS Distance
FROM VECTOR_SEARCH(
    TABLE = dbo.ProductReview AS r,
    COLUMN = ReviewVector,
    SIMILAR TO = @searchVector,
```

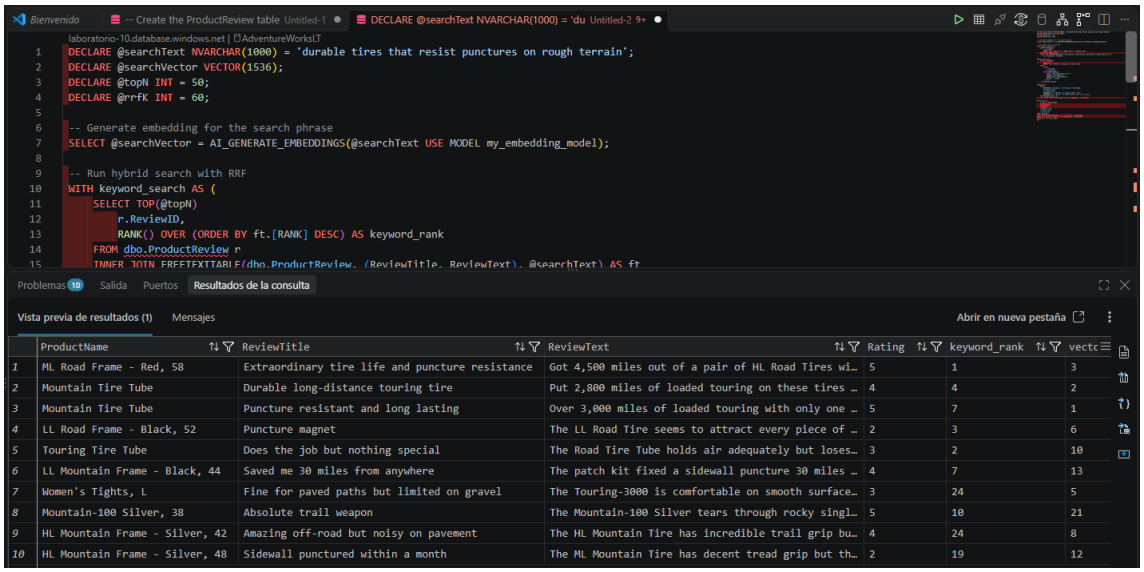
	ProductName	ReviewTitle	ReviewText	Rating	Category	Distance
1	HL Touring Frame - Yellow, 50	Essential for commuter visibility	I mounted this taillight on my seatpost and it is…	5	Touring Frames	0,4116618
2	Mountain Bike Socks, M	Visible and ventilated for night riding	The Sport-100 in black has reflective accents tha…	5	Socks	0,4413521
3	Sport-100 Helmet, Red	Bright and visible for safety	Chose the Sport-100 Red specifically for visibili…	5	Helmets	0,4628318
4	HL Touring Frame - Yellow, 54	Game changer for predawn rides	My shift starts at 5 AM so I ride in pitch darkne…	5	Touring Frames	0,4672518
5	HL Touring Frame - Yellow, 50	Bright and reliable tail light	The flashing mode on this taillight is extremely …	4	Touring Frames	0,4912042
6	HL Touring Frame - Yellow, 54	Lights up the darkest rural roads	These dual-beam headlights turned sketchy nightti…	5	Touring Frames	0,5012041
7	HL Touring Frame - Blue, 46	Survives any weather condition	The weatherproof headlights survived a month of r…	5	Touring Frames	0,5215025
8	HL Touring Frame - Blue, 46	Survived a torrential downpour	Got caught in a torrential downpour on a 50-mile …	5	Touring Frames	0,5684858
9	Taillights - Battery-Powered	Crossed three states on this bike	Completed a two-week tour with 40 pounds of gear …	5	Lights	0,5852853
10	HL Touring Frame - Yellow, 50	Mounting bracket cracked in cold weather	The taillight itself is bright and visible but th…	2	Touring Frames	0,6045794
11	LL Mountain Frame - Black, 44	Saved me 30 miles from anywhere	The patch kit fixed a sidewall puncture 30 miles …	4	Mountain Frames	0,6085152
12	Taillights - Battery-Powered	Reliable but gearing could be lower	The Touring-1000 Blue handles loaded climbing wel…	4	Lights	0,6120379
13	HL Touring Frame - Blue, 46	Battery indicator is wildly inaccurate	The light itself is great — bright, waterproof, g…	3	Touring Frames	0,6213397
14	Sport-100 Helmet, Black	Great starter helmet for new riders	The Sport-100 Blue was the first piece of cycling…	5	Helmets	0,6300746
15	ML Road Frame-W - Yellow, 48	Solid daily commuter	Switched from the bus to cycling 9 miles to my of…	4	Road Frames	0,6307921
16	ML Road Frame - Red, 58	Confidence in wet corners	The HL Road Tire channels water effectively and I…	4	Road Frames	0,6319909

Comprobamos los vectores por los meta data



Combine full-text and vector search with hybrid search

Combinamos ambas del scrip que nos aportan es necesario eliminar el topN=50 porque da error



Compare the three search approaches

Realizamos una búsqueda de texto completa

```
laboratorio-10.database.windows.net | AdventureWorksLT
1  -- Full-text search only
2  SELECT TOP 5
3      p.Name AS ProductName,
4      r.ReviewTitle,
5      r.ReviewText,
6      r.Rating,
7      ft.[RANK] AS FullTextRank
8  FROM dbo.ProductReview r
9  INNER JOIN SalesLT.Product p ON r.ProductID = p.ProductID
10 INNER JOIN FREETEXTTABLE(dbo.ProductReview, (ReviewTitle, ReviewText),
11     'comfortable bike for long weekend rides with the family') AS ft
12     ON r.ReviewID = ft.[KEY]
13 ORDER BY ft.[RANK] DESC;
14 GO
```

Problemas 2 | Salida | Puertos | **Resultados de la consulta**

Vista previa de resultados (1) | Mensajes | Abrir en

	ProductName	ReviewTitle	ReviewText	Rating	FullTextRank
1	Water Bottle - 30 oz.	Stays cold for hours on long rides	We do 3-hour family rides along the river trail e…	5	108
2	Water Bottle - 30 oz.	Perfect family hydration solution	Bought four of these 30 oz water bottles — one fo…	5	108
3	Long-Sleeve Logo Jersey, L	Matching jerseys for the whole family	Ordered the Long-Sleeve Logo Jersey in different …	5	92
4	Road-550-W Yellow, 44	Great family weekend bike	Bought the Mountain-500 Black for Saturday rides …	4	88
5	ML Road Frame-W - Yellow, 48	Great first road bike	Transitioned from a hybrid to the Road-650 Black …	4	76

Hacemos una búsqueda de vector con el mismo parámetro

Bienvenido | -- Create the ProductReview table Untitled-1 | -- Vector search only Untitled-2

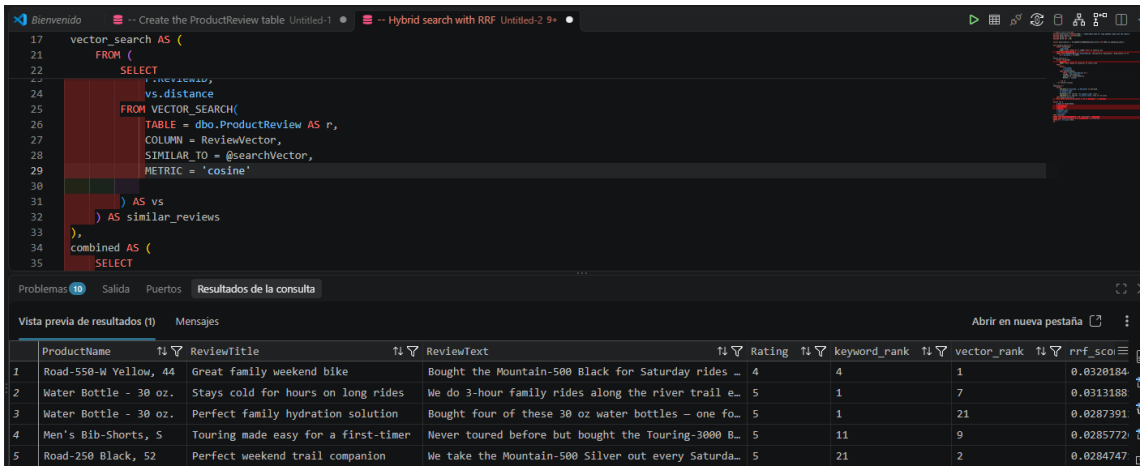
```
laboratorio-10.database.windows.net | AdventureWorksLT
1  -- Vector search only
2  DECLARE @searchText NVARCHAR(1000) = 'comfortable bike for long weekend rides with the family';
3  DECLARE @searchVector VECTOR(1536);
4
5  SELECT @searchVector = AI_GENERATE_EMBEDDINGS(@searchText USE MODEL my_embedding_model);
6
7  SELECT
8      p.Name AS ProductName,
9      r.ReviewTitle,
10     r.ReviewText,
11     r.Rating,
12     vs.distance AS Distance
13 FROM VECTOR_SEARCH(
14     TABLE = dbo.ProductReview AS r,
15     COLUMN = ReviewVector,
```

Problemas | Salida | Puertos | **Resultados de la consulta**

Vista previa de resultados (1) | Mensajes | Abrir en nueva pestaña

	ProductName	ReviewTitle	ReviewText	Rating	Distance
1	Road-550-W Yellow, 44	Great family weekend bike	Bought the Mountain-500 Black for Saturday rides …	4	0,34975099563598633
2	Road-250 Black, 52	Perfect weekend trail companion	We take the Mountain-500 Silver out every Saturda…	5	0,4646100401878357
3	HL Mountain Rear Wheel	Sunday ride perfection	The Road-550-W is my go-to for Sunday morning rid…	5	0,46839362382888794
4	Men's Sports Shorts, L	Capable mid-range touring platform	The Touring-2000 handled a 400-mile charity ride …	4	0,47596853971481323
5	Touring-Panniers, Large	Built for the long haul	Took the Touring-1000 Yellow on a 600-mile ride a…	5	0,4950132369995117

Usamos la búsqueda hidrida


Bienvenido
-- Create the ProductReview table Untitled-1
-- Hybrid search with RRF Untitled-2 9+
vector_search AS (
FROM (
SELECT
vs.distance
FROM VECTOR_SEARCH(
TABLE = dbo.ProductReview AS r,
COLUMN = ReviewVector,
SIMILAR_TO = @searchVector,
METRIC = 'cosine'
) AS vs
) AS similar_reviews
),
combined AS (
SELECT
Problemas 10
Salida
Puertos
Resultados de la consulta
Vista previa de resultados (1)
Mensajes
Abrir en nueva pestaña
ProductName
ReviewTitle
ReviewText
Rating
keyword_rank
vector_rank
rrf_scor
1 Road-550-W Yellow, 44 Great family weekend bike Bought the Mountain-500 Black for Saturday rides … 4 4 1 0.0320184
2 Water Bottle - 30 oz. Stays cold for hours on long rides We do 3-hour family rides along the river trail e… 5 1 7 0.0313188
3 Water Bottle - 30 oz. Perfect family hydration solution Bought four of these 30 oz water bottles — one fo… 5 1 21 0.0287391
4 Men's Bib-Shorts, S Touring made easy for a first-timer Never toured before but bought the Touring-3000 B… 5 11 9 0.0285772
5 Road-250 Black, 52 Perfect weekend trail companion We take the Mountain-500 Silver out every Saturda… 5 21 2 0.0284747